

Trace Tables

Project: Transparent Book Recommender

1. Recall and choose your starting point

Try the recall before opening notes. Then use a different colour or a labelled second line to correct your explanation.

Branching Logic

Without opening the old lesson, explain one key idea from Branching Logic. Give an example and a mistake that the idea helps you avoid.

My remembered idea, example and correction

Precision in Algorithms

Without opening the old lesson, explain one key idea from Precision in Algorithms. Give an example and a mistake that the idea helps you avoid.

My remembered idea, example and correction

2. Practise with fading help

Use W2: Trace before running for Loops and trace tables. Record the answer once; the lesson and workbook refer to the same attempt.

Given inputs · Procedure I chose · Middle steps · Result and check

Given inputs: _____

Procedure I chose: _____

Middle steps: _____

Result and check: _____

The step I completed without help, and the hint I needed

3. Individual reasoning check

Use the worked case for Loops and trace tables. Proposed explanation: “The stop value is included”.

Decide whether this explanation is supported. Point to the step or supplied fact that decides; explain your repair if needed.

My explanation before partner discussion

4. Independent transfer case

Independent case: Build a repeated addition independently

A box starts with 5 counters. Each round adds 2 counters. Use rounds = 3, an initialized total and for turn in range(rounds). Print the final total once after the loop. Trace the totals and predict fresh runs with rounds 0, 1 and 3. Assume rounds is a nonnegative integer.

My independent explanation, trace and limits

Attempt this case before feedback. Record any help. Earlier examples below are further practice; a case already practised with feedback cannot establish fresh transfer.

5. Your learning target

Start score=1; add 4; if score>3 subtract 2; double score. Record score after every operation. A student evaluates the condition using the initial value. Locate the divergence.

6. Solve the given case

Set score=2; add 3; if score>4 subtract 1; double score.

Trace every value.

Show your trace, calculation or evidence

Write or draw your reasoning and evidence.

7. Check a changed case

Set score to 2. Add 3, then double the result. Record the score after each operation. What changes if you double before adding?

My independent answer and reason

Write or draw your reasoning and evidence.

Week 4 Test and reflect

8. Design a revealing test

Create a changed input, boundary case or missing-information case. Write the expected answer or safe response before testing.

Test record · My evidence

Test record: Changed input or condition

My evidence: _____

Test record: Expected answer or fallback

My evidence: _____

Test record: Observed result

My evidence: _____

Test record: Why the result occurred

My evidence: _____

9. Project checkpoint

Trace each book's genre, length, difficulty, and total score before sorting.

The project change and evidence I will keep

Write or draw your reasoning and evidence.

10. Self check

I can explain the mechanism. I can show evidence. I can handle a changed case. I can name a limit. Mark each as not yet, with help or independently, and attach supporting work.

Teacher review _____ Next step _____

11. Transfer practice from the original programme

Start score=1; add 4; if score>3 subtract 2; double score. Record score after every operation. A student evaluates the condition using the initial value. Locate the divergence.

My independently reasoned answer

Record support used: none, hint or teacher explanation.

12. W1 Explain the range

List the values supplied by range(2, 5) and explain why 5 is absent. How many times does its body run?

My reasoning and evidence

13. W2 Trace before running

Copy the worked script. Change the range to range(2, 5). Record turn, total before and total after each iteration. Predict the single output, then run and compare.

My reasoning and evidence

14. W3 Repair the reset

For the faulty script with range(1, 4), record total after each addition. Explain the wrong final output and remove only the faulty reset.

My reasoning and evidence

15. W4 Build a repeated addition independently

A box starts with 5 counters. Each round adds 2 counters. Use rounds = 3, an initialized total and for turn in range(rounds). Print the final total once after the loop. Trace the totals and predict fresh runs with rounds 0, 1 and 3. Assume rounds is a nonnegative integer.

My reasoning and evidence

16. W5 Check a contract limitation

In your counter script, `range(-2)` performs no iterations. Does returning the initial 5 make negative rounds valid? Explain what the script checks and what it assumes.

My reasoning and evidence

17. Feedback, return check and learning record

Evidence · My record

Evidence: Worked alone / hint / model

My record: _____

Evidence: First step I repaired and why

My record: _____

Evidence: New case checked later; date and result

My record: _____

Evidence: What I can now explain without notes

My record: _____

Evidence: One remaining question

My record: _____

Revisit this idea with your teacher after a delay. Familiarity is different from explaining and using it on a changed case.

18. Pictorial case practice: Trace Tables

A fictional reading-token jar starts with one token and gains three per round.

Python: `total=1; for turn in range(4): total=total+3; after the loop print(total).`

The update is inside the loop; print is outside.

Loop totals after each iteration

	turn	total after update
First	0	4
Second	1	7
Third	2	10
Fourth	3	13

Read a cell using both its row and column labels.

Trace Tables

Start with the supplied facts

Follow each arrow. Ask which fact allows the next step.



1. Initial state

Before the loop, total=1.



2. Iteration values

range(4) supplies turn values 0,1,2,3.



3. Updated totals

After each turn total is 4,7,10,13.



4. After the loop

Print 13 once, after all four updates.

CHANGE ONE THING

Use range(0) instead of range(4).

Explain the changed result using the matching lesson.

Move print inside the original loop. What sequence is displayed?

My picture-based explanation and changed-case prediction

Support used: alone / hint / worked example. This record is practice evidence.

Project connection: Trace recommender loops with current item, eligibility, score and current best so each update can be explained.

19. Project portfolio milestone

Project: Transparent Book Recommender

Current milestone: Frame the need. Name the user, the problem, allowed inputs and a safe simple starting method.

Trace recommender loops with current item, eligibility, score and current best so each update can be explained.

Evidence to keep · My record

Evidence to keep: This week’s input and deciding step

My record: _____

Evidence to keep: Prediction and actual result (or labelled paper trace)

My record: _____

Evidence to keep: One change since the previous checkpoint and why

My record: _____

Evidence to keep: One error or limit and the next test

My record: _____

Explain what this week adds to your project. Reuse last week’s record rather than starting again.

20. Supported practice and learning evidence

Use this supported practice once, in the browser or on paper. Keep the reserved independent assessment for a separate first attempt before teacher feedback.

A fictional reading-token jar starts with one token and gains three per round.

Practice 1: explain the deciding evidence

Which labelled step matches this detail? Print 13 once, after all four updates.

1. Iteration values
2. Updated totals
3. After the loop
4. Initial state

My selected conclusion, deciding fact and explanation

Hint if needed: Read the steps in the pictorial case. Match the supplied detail to the stage that performs or records it.

After your attempt, compare with the supported explanation: After the loop Print 13 once, after all four updates. The turn value labels an iteration; it is not the number added. The fixed update adds three each time. Four iterations take the total from 1 to 13. Keeping print outside the loop produces one final display rather than four intermediate displays. Initialise total before the loop so a zero-iteration run still has a defined value.

Practice 2: explain the deciding evidence

Changed-case practice: Use `range(0)` instead of `range(4)`. Which conclusion is supported by the supplied facts?

1. The loop body never runs and the printed total is 1.
2. This one example guarantees correct results for every future input.
3. We can decide the result without examining the changed input or the procedure.

My selected conclusion, deciding fact and explanation

Hint if needed: Use the changed condition and the deciding step in the pictorial trace. The answer must say what follows from those facts.

After your attempt, compare with the supported explanation: The loop body never runs and the printed total is 1. The iterable contains no turn values; the initial assignment still happened.

Repair a missing building block

Review Branching Logic (ai-2/3). Goal: Trace the selected branch in an if elif else chain. Short practice: Give expected labels for -1,0,10,11 and identify the equality boundary for Short.. Return to this week and explain what you changed.

Review Precision in Algorithms (ai-2/1). Goal: Translate a short paper calculation into a Python script. Short practice: If `print(total)` moves before `total=total+2`, what gets printed for the original input?. Return to this week and explain what you changed.

Predict, run and debug

The browser lab lets you edit the supported Python examples already included in this lesson. Predict first, run, inspect the actual output or error, change one input, then run again. Keep code, predictions, actual results and a reasoned revision together. On paper, label your result as a trace rather than an execution.

Keep your completed work

Use Download completed work in the hosted lesson to export your answers, reflections, practice feedback and code-run evidence. Open the HTML copy to read or print it. Drafts stay in this browser when storage is available; clear drafts on a shared device.

What I repaired, the evidence I kept and the next test
